Enhancement Three Narrative

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1. Briefly describe the artifact. What is it? When was it created?

   The artifact is my InputOutput.cpp, InvestmentAccountInfo.cpp, and main.cpp files in my Investment Calculator Application that I created in CS-210 Programming Languages, back in February 2025.

   There are also the additional files that needed to be created in order to fully enhance these artifacts. The additional files that were created are UserDatabase.cpp and UserDatabase.h.

2. Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

   The main reason that I included these as my artifact for my ePortfolio is because there was no database for this application and I felt that it needed a simple and local way of storing user data so users can reload their own investment information if they want to come back and mess with the different attributes.

   This will show my ability to successfully incorporate community made libraries and use them inside existing programs. This requires reading documentation from the developer and being able to problem solve issues that arise.

The specific components that will showcase my skills and abilities in software development is how and what I used to add a database to my C++ application. I decided to use Json as my database, which is not natively supported as a database type for C++. I found a library and project on GitHub which makes it possible to add Json support in C++. It's called JSON for Modern C++ by Niels Lohmann which can be found here: https://github.com/nlohmann/json

3. Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

   I did meet the course outcomes that I planned to meet with this enhancement.

   I do have an update but in the sense that I feel that this enhancement alone is meeting all course outcomes in one way or another. I have to redesign what I have already done in my first enhancement in order to successfully incorporate Json into the application. This also meets what was needed in the enhancement two as well, with having to change algorithms and data structures in order to incorporate the Json database and save and load user data.

4. Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

   I felt extremely confident at first when starting the process of enhancing and modifying this artifact. I quickly found out that chasing down issues with saving and

loading user data was like going down a rabbit hole. I at first figured I could just wing it and quickly add this database feature with no issues. After feeling like I made more of a mess of the code and running into a slight bit of confusion, I went back to the drawing board and back to the basics of computer science. Looked through what I had already and focused on writing tons of comments throughout my files to help facilitate me keeping on track and not jumping around. I discovered nearly a year ago that I do this thing called feature-creep where I tend to add additional things before I even need them which causing unforeseen issues and bugs when I test the program. I stick with one feature and if I want to add something I turn it into a comment, so I don’t forget.